

Evaluating Large Language Models for Cross-Lingual Retrieval

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Motivation

- Cross-lingual IR (CLIR) based on LLMs remains underexplored.
- Prior work [1] relies on machine translation (MT) → costly, prone to error propagation.
- We evaluate LLMs on two CLIR benchmarks (CLEF2003, CIRAL), reranking paradigms (pairwise, listwise), and different retrieval and reranking models.

Research Questions

- **RQ1:** How do recent multilingual bi-encoders and translation-based sparse retrieval compare in their performance? (see paper)
- **RQ2:** Do LLM-based rerankers improve CLIR performance across high-resource and low-resource languages?
- **RQ3:** How do pairwise and listwise rerankers influence CLIR performance?
- **RQ4:** What is the impact of document length on reranking in CLIR?

Retrieval and Reranking Models

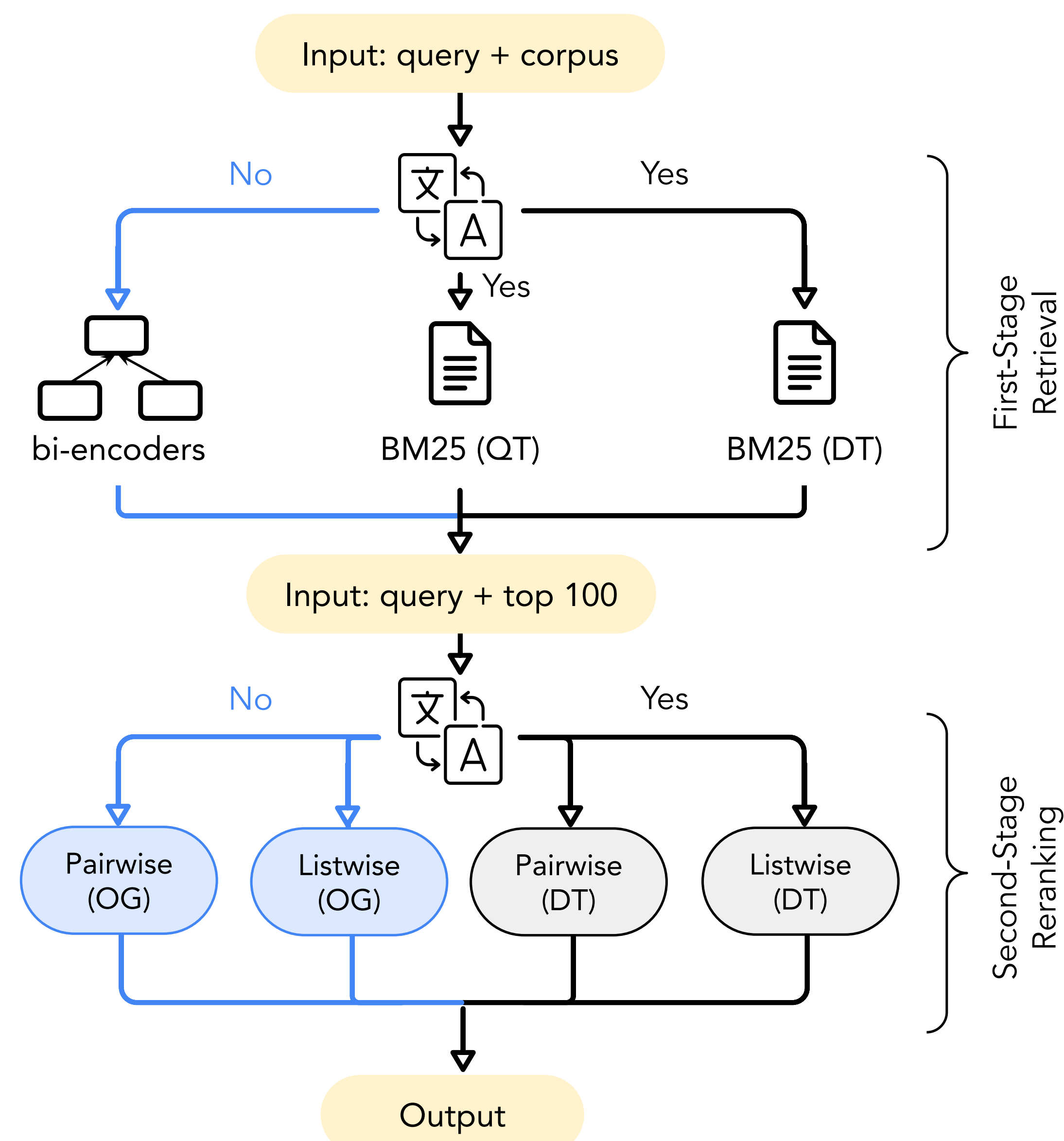
First-Stage Retrieval Models

- BM25 + Machine Translation.
- Bi-encoders: M3, mGTE, E5, RepLLaMa, NV-Embed-v2.

Second-Stage Reranking Models

- Listwise [2]: RankZephyr, RankGPT3.5, RankGPT4.1.
- Pairwise [3]: Llama-3.1-8B-Instruct, Aya-101.

Cross-Lingual Retrieval Pipeline



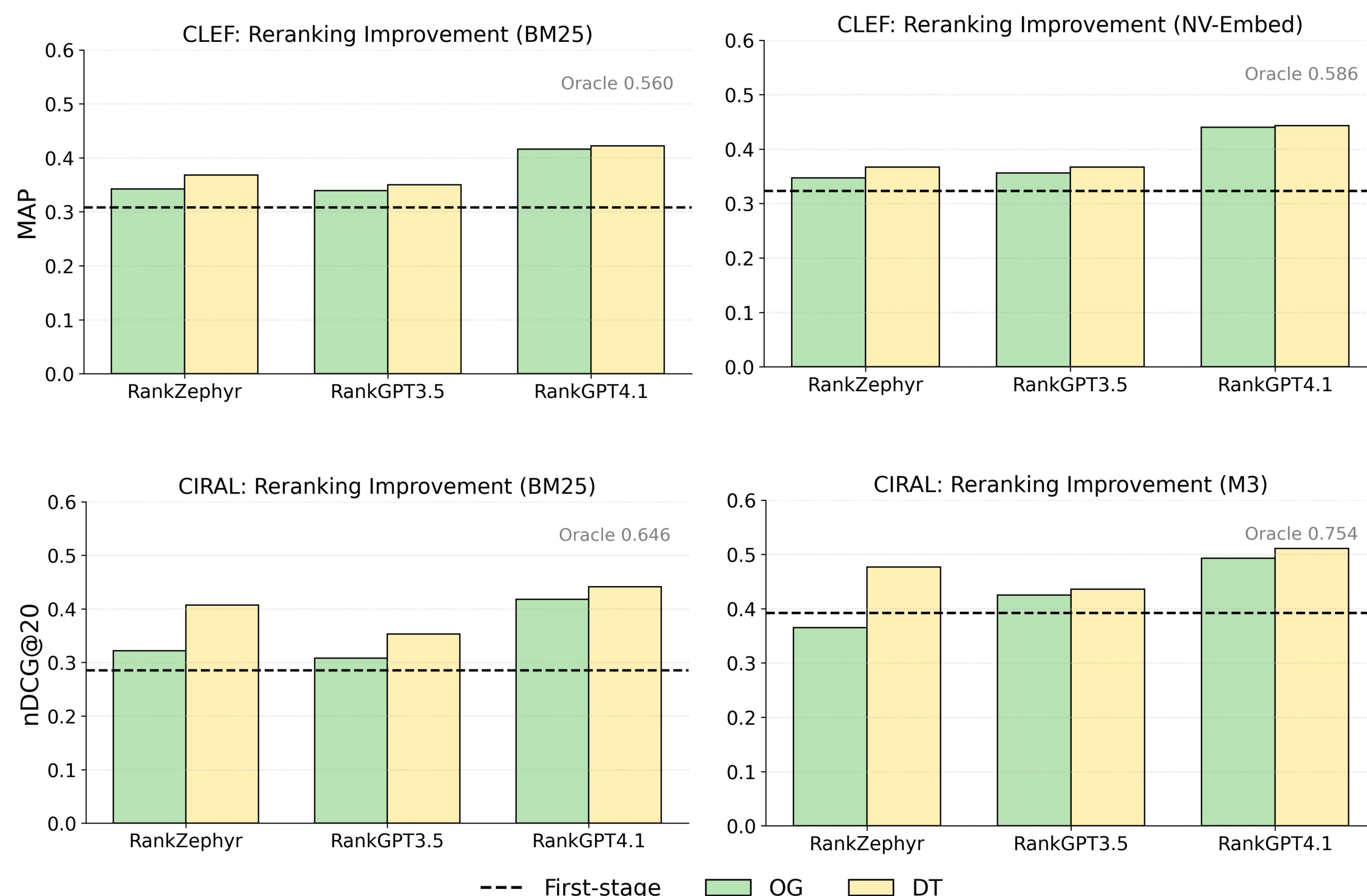
Results and Discussion (RQ2)

Second-Stage Listwise Reranking

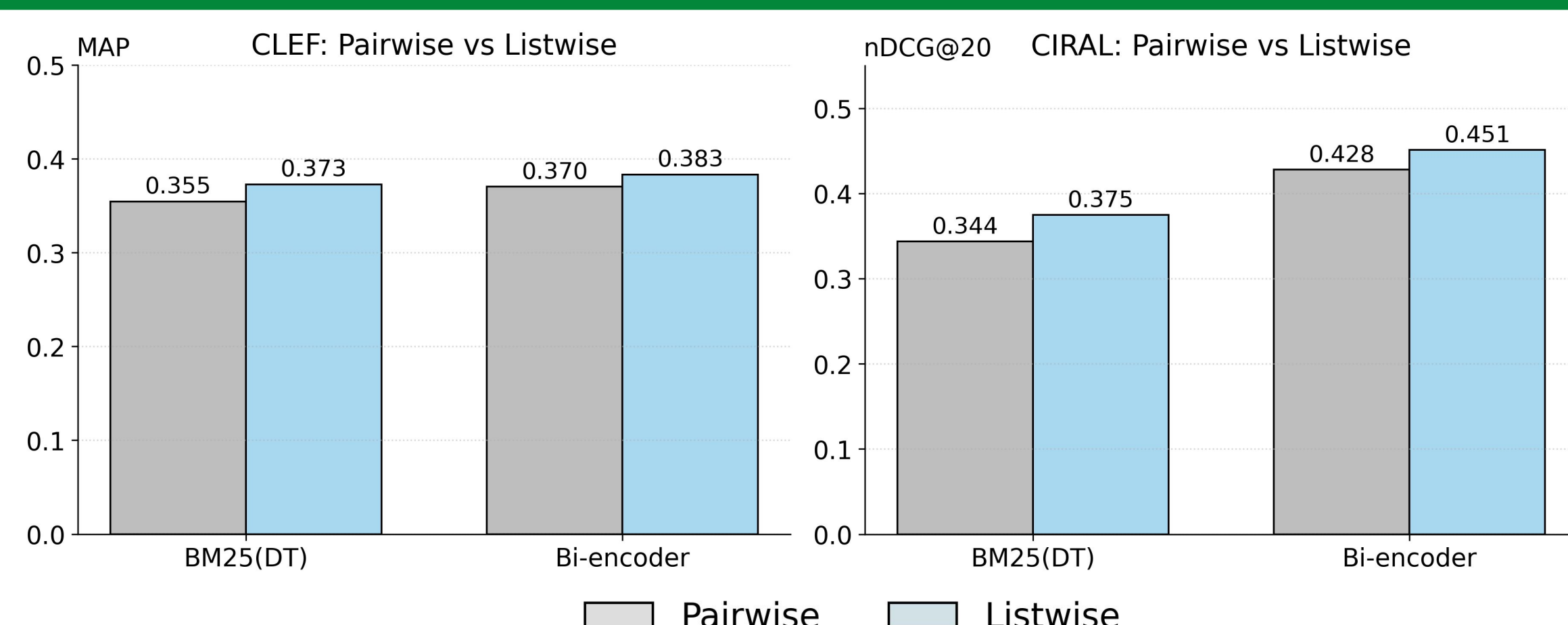
- Consistent across both datasets, better retrieval yields better reranking. (RQ2)
- The benefits from DT appears to diminish, as models achieve stronger performance.

The Glass Ceiling of Reranking

- “Ceiling Effect”: even state-of-the-art rerankers struggle to approach the upper bound.

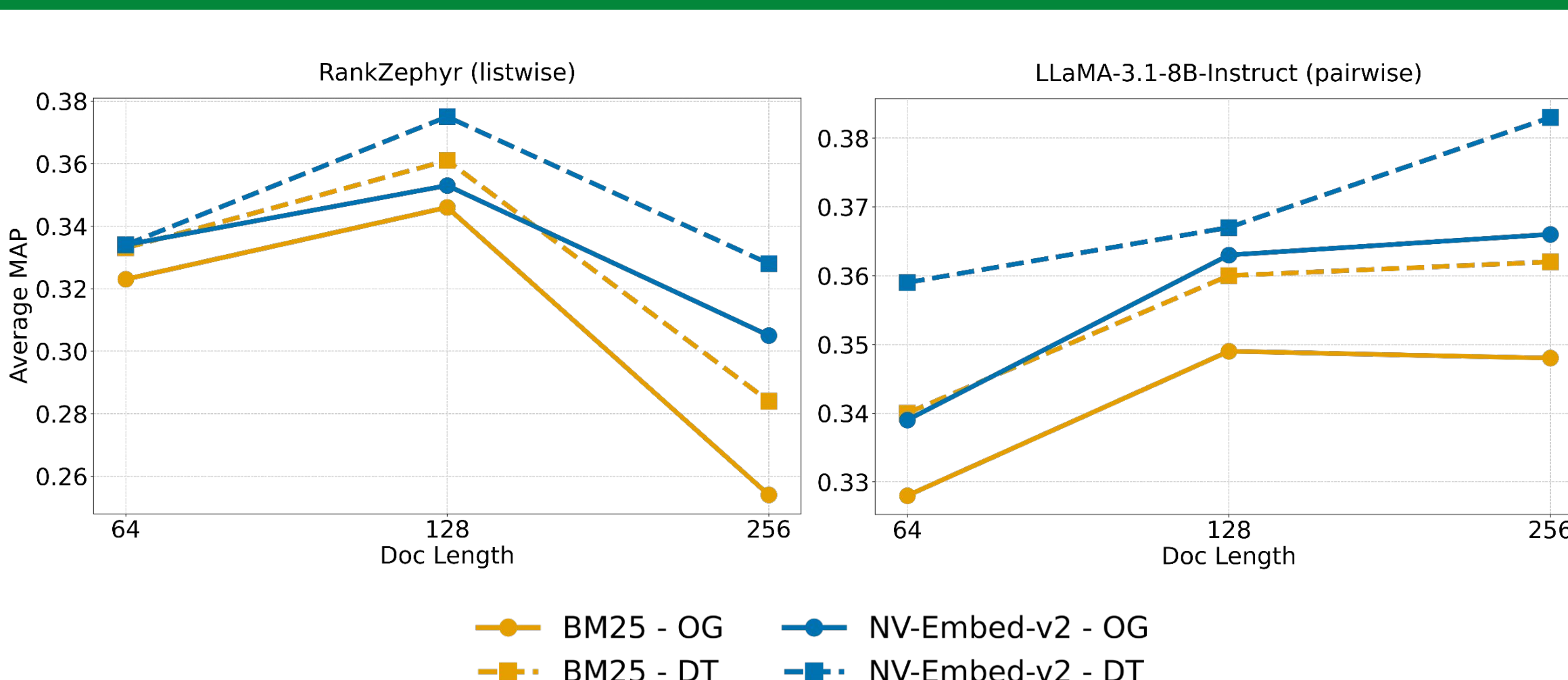


Pairwise Reranking (RQ3)



- All pairwise reranking models improve input rankings.
- Models perform competitive to listwise rerankers.

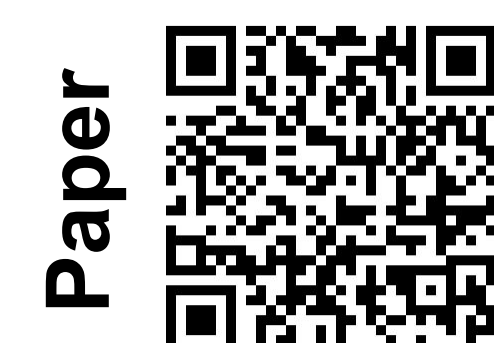
Impact of Document Length (RQ4)



- Listwise rerankers perform worse on long inputs.
- Pairwise rerankers are more robust to longer inputs

Conclusion

- Improved retrieval results translates to better reranking results, the benefits of MT diminish with stronger listwise rerankers.
- Pairwise rerankers, based on instruction-tuned LLMs, perform competitive to listwise rerankers trained on retrieval supervision.
- Performance varies with data conditions — reranking performance is sensitive to document lengths and language pairs.



References

- [1] Adeyemi, Mofetoluwa, et al. "Zero-Shot Cross-Lingual Reranking with Large Language Models for Low-Resource Languages." In Proceedings of ACL 2024.
- [2] Xueguang Ma et al. "Zero-shot listwise document reranking with a large language model." Preprint, arXiv:2305.02156. 2023.
- [3] Qin, Zhen, et al. "Large Language Models are Effective Text Rankers with Pairwise Ranking Prompting." In Findings of NAACL 2024.